



Designing a Next-Generation Electric Bus with High Level of Service Corridor for Regional Mobility: A Case Study

Samera Ibrahim Kadum Aladdal^{1*}, Sawsan Kareem Kadhim²

¹ Department of Regional Planning, Faculty of Urban Planning, University of Kufa, 54001 Najaf, Iraq

² Department of Civil Engineering, College of Engineering, University of Kufa, 54001 Najaf, Iraq

* Correspondence: Samera I. Aladdal (samera.aladdal@uokufa.edu.iq)

Received: 12-11-2025

Revised: 05-18-2026

Accepted: 05-25-2026

Citation: S. I. K. Aladdal and S. K. Kadhim, “Designing a next-generation electric bus with high level of service corridor for regional mobility: A case study,” *Int. J. Transp. Dev. Integr.*, vol. 10, no. 2, pp. 454–477, 2026. <https://doi.org/10.56578/ijtdi100209>.



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Abstract: This study develops an integrated planning and operational framework for a next-generation electric bus with high level of service (Electric-BHLS) corridor along the Najaf–Al-Manathira–Al-Meshkhab axis in Iraq. The corridor represents a strategically important urban–rural mobility corridor characterized by rapid demographic growth, fragmented public transport services, congestion, environmental degradation, and increasing dependence on informal low-capacity vehicles. Unlike conventional electric bus operations, the proposed Electric-BHLS model combines high-service operational characteristics—including adaptive service frequency, intelligent transport systems (ITS)-based fleet control, hybrid priority lanes, opportunity charging systems, and real-time operational management—with full electric propulsion and regional accessibility planning. The methodological framework integrates engineering analysis, Geographic Information System (GIS)-based spatial accessibility assessment, operational modeling, and generalized cost optimization. Empirical calibration is based on 2024 field surveys, passenger interviews, Global Positioning System (GPS) based travel-time measurements, institutional datasets, and corridor infrastructure assessments. The proposed system includes articulated electric buses, pantograph opportunity-charging infrastructure, centralized Operations Control Center (OCC) management, smart passenger information systems, and a hierarchical station structure designed to improve operational reliability and multimodal integration. The results demonstrate substantial operational, environmental, and spatial improvements compared with the existing transport system. The optimized Electric-BHLS configuration reduces generalized transport cost by 27%, decreases average passenger waiting time by 61%, and lowers carbon dioxide (CO₂) emissions by approximately 29%. Corridor passenger capacity increases from approximately 15,000 to 36,000 passengers per day, while average operating speed improves from 22 km/h to 35 km/h through ITS-supported operational control and selective priority measures. GIS analysis further indicates accessibility gains of 24% in urban areas and 38% in rural catchment zones, improving access to employment, education, healthcare, and regional services. Beyond technical performance, the study evaluates governance, financial feasibility, operational risk, and long-term implementation constraints within the Iraqi context. A phased Design–Build–Operate–Maintain (DBOM) Public–Private Partnership (PPP) framework and a unified corridor governance authority are proposed to support institutional coordination and long-term operational sustainability. The study concludes that Electric-BHLS represents a scalable and economically viable mobility model capable of supporting sustainable regional development and transport modernization in Iraq and comparable developing-country contexts.

Keywords: Electric-BHLS; Sustainable mobility; Intelligent transport systems; Regional accessibility; Generalized cost optimization; Public transport planning; Electric transit; Iraq

1 Introduction

Transport systems in developing countries are increasingly confronted with complex mobility challenges resulting from rapid urbanization, population growth, informal transport expansion, environmental degradation, and insufficient infrastructure investment [1, 2]. These challenges are particularly severe in medium-sized and rapidly expanding urban regions where transport demand has grown faster than the capacity of existing public transport systems. In many developing contexts, mobility networks are characterized by fragmented institutional structures, weak

operational coordination, low-capacity vehicle fleets, and limited integration of intelligent transport technologies [3]. Consequently, urban and regional corridors frequently experience chronic congestion, unreliable travel times, declining service quality, and rising environmental pressures.

Iraq reflects many of these structural transport challenges. Decades of underinvestment, rapid demographic growth, and heavy dependence on private and informal transport modes have generated increasing pressure on regional mobility systems [4]. Public transport services in many Iraqi cities remain dominated by small-capacity minibuses and taxis operating under weak regulatory and operational frameworks, limiting both efficiency and accessibility [5]. These conditions are particularly visible in Najaf Governorate, one of Iraq's most important religious, economic, and regional centers. The governorate experiences substantial daily mobility flows associated with employment, education, commercial activities, agricultural production, and religious tourism. The Najaf–Al-Manathira–Al-Meshkhab corridor constitutes one of the most strategically significant regional axes within the governorate, linking dense urban districts with semi-urban and rural settlements while simultaneously supporting seasonal pilgrimage movement and agricultural logistics.

Despite its strategic importance, the corridor currently suffers from severe operational and infrastructural deficiencies. These include peak-hour congestion, unreliable travel times, fragmented transport supply, inadequate passenger facilities, weak service integration, and increasing environmental impacts caused by aging vehicle fleets [6]. Existing public transport services depend largely on informal small-capacity vehicles operating in mixed traffic conditions without centralized operational control or intelligent traffic-management systems. The absence of a structured high-capacity public transport backbone has reduced regional accessibility, increased user travel costs, weakened spatial connectivity, and constrained long-term socio-economic development across both urban and rural communities.

In response to these challenges, high-quality bus-based transit systems have emerged internationally as cost-effective and operationally flexible alternatives to heavy rail and fully segregated Bus Rapid Transit (BRT) systems. Among these approaches, bus with high level of service (BHLS) systems have attracted growing attention because of their ability to combine operational efficiency, service reliability, enhanced passenger experience, and moderate infrastructure requirements [7, 8]. Unlike conventional bus operations, BHLS systems incorporate integrated operational features such as service-priority measures, centralized operational management, intelligent transport systems (ITS), real-time passenger information, and enhanced station design. Compared with fully segregated BRT systems, BHLS systems also provide greater adaptability and financial feasibility for medium-sized cities and developing regional corridors where financial resources, roadway space, and implementation capacity remain constrained [9].

The global transition toward transport electrification has further expanded the strategic role of BHLS systems within sustainable mobility planning. Electrified high-service bus systems offer substantial advantages in terms of energy efficiency, operational performance, noise reduction, and greenhouse-gas mitigation [10, 11]. Recent international experiences in cities such as Lyon, Amsterdam, Shenzhen, and Tashkent demonstrate that the integration of electric propulsion, fast-charging systems, and ITS-based fleet management can significantly improve both operational and environmental performance [12, 13]. Nevertheless, much of the existing literature examines either electric bus technologies or conventional BRT/BHLS operations separately, while limited research addresses fully integrated next-generation electric bus with high level of service (Electric-BHLS) systems within regional corridors in developing-country environments.

This gap is particularly evident in Iraq and the broader Middle Eastern context, where research on sustainable regional transit systems remains limited. Existing studies in Iraq primarily focus on conventional road transport problems, traffic congestion, or isolated urban transport interventions [14, 15]. Few studies examine the integration of electric mobility, ITS, spatial accessibility analysis, operational optimization, and governance frameworks within a unified regional transit model. Moreover, existing research often treats electrification, operational management, and accessibility planning as separate technical domains rather than interconnected dimensions of regional mobility transformation. Limited attention has also been given to the operational feasibility of Electric-BHLS systems under conditions characterized by institutional fragmentation, infrastructure limitations, and mixed urban–rural mobility patterns.

Accordingly, this study develops an integrated planning and operational framework for an Electric-BHLS corridor along the Najaf–Al-Manathira–Al-Meshkhab axis. The proposed model combines electric propulsion, ITS-supported fleet management, adaptive service frequency, hybrid priority-lane strategies, Geographic Information System (GIS)-based accessibility planning, and generalized transport-cost optimization within a unified analytical framework. Unlike conventional electric bus operations, the proposed Electric-BHLS concept emphasizes coordinated operational control, real-time fleet monitoring, multimodal integration, service-quality optimization, and spatial accessibility enhancement to achieve high levels of reliability, efficiency, and regional connectivity. Attention is also given to spatial equity through improving accessibility for underserved rural and semi-urban communities.

The methodological framework integrates engineering analysis, spatial accessibility assessment, operational

modeling, and sustainability evaluation. Empirical calibration is based on field surveys, GPS-based travel-time measurements, passenger interviews, and institutional datasets collected during 2024. The study evaluates operational performance, environmental impacts, accessibility improvements, financial feasibility, and institutional implementation requirements within the Iraqi context. In addition, the research proposes a phased DBOM–PPP structure and a unified corridor governance authority to support long-term operational coordination and institutional sustainability.

The study contributes to existing literature in several important ways. First, it develops one of the first integrated Electric-BHLS planning frameworks for Iraq and comparable developing-country environments. Second, it combines engineering design, operational optimization, GIS-based spatial analysis, and governance assessment within a unified methodological structure. Third, it introduces a corridor-scale perspective that links sustainable mobility planning with regional accessibility, socio-economic integration, spatial equity, and long-term transport modernization. Finally, the study provides a transferable framework for the deployment of sustainable electric transit systems in rapidly urbanizing regions characterized by fragmented transport systems and infrastructure constraints.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical foundations and previous studies related to BHLS systems, electric public transport, ITS integration, and transport optimization models. Section 3 presents the study area and the operational characteristics of the Najaf–Al-Manathira–Al-Meshkhab corridor. Section 4 develops the proposed Electric-BHLS system design and operational framework. Section 5 outlines the methodological framework and optimization model. Section 6 presents the results and discussion, including operational, environmental, and accessibility outcomes. Finally, Section 7 concludes the study and presents policy recommendations and future research directions.

2 Theoretical Framework and Previous Studies

2.1 Evolution of High-Service Bus Systems

The development of high-capacity and high-quality bus transit systems has become a central component of sustainable urban and regional mobility strategies worldwide. Conventional bus systems have historically suffered from operational inefficiency, low reliability, congestion exposure, and weak passenger attractiveness, particularly in rapidly urbanizing cities [16]. In response, several transit models have emerged to improve public transport performance while maintaining lower infrastructure costs than heavy rail systems.

BRT systems represented one of the earliest large-scale attempts to achieve rail-like operational performance through segregated corridors, platform-level boarding, centralized operations, and service-priority mechanisms [8]. Cities such as Curitiba, Bogotá, and Guangzhou demonstrated that BRT systems could significantly improve passenger mobility, reduce congestion, and support sustainable urban growth [17]. However, full BRT systems often require extensive infrastructure investment, large right-of-way allocation, and substantial urban reconstruction, making implementation difficult in medium-sized and infrastructure-constrained cities.

As an alternative, BHLS systems emerged in Europe as a more flexible and adaptable operational model [7]. BHLS systems maintain several advanced operational characteristics associated with BRT—such as intelligent fleet management, high service frequency, passenger-information systems, operational control centers, and selective priority measures—while allowing partial integration with existing urban road networks [18]. This flexibility has enabled BHLS systems to operate effectively within medium-density cities and mixed urban–regional environments where fully segregated BRT infrastructure is financially or spatially infeasible.

Unlike conventional bus operations, BHLS systems emphasize operational quality, service regularity, passenger comfort, and real-time system management. Core characteristics include centralized Operations Control Center (OCC), adaptive service scheduling, dedicated or semi-priority lanes, ITS, platform-level accessibility, multimodal integration, and enhanced station infrastructure [19, 20]. Figure 1 illustrates the integrated operational and analytical framework of the proposed Electric-BHLS system. The figure compares the operational characteristics of conventional bus systems, BRT systems, and Electric-BHLS systems, while also presenting the integrated analytical structure incorporating ITS architecture, operational optimization, accessibility analysis, and centralized operational management.

Recent studies indicate that BHLS systems can achieve significant improvements in travel-time reliability, passenger satisfaction, operational efficiency, and network flexibility without the full capital intensity associated with metro or heavy BRT systems [21]. Moreover, the operational adaptability of BHLS systems makes them particularly suitable for developing-country corridors characterized by fragmented urbanization patterns and mixed urban–rural mobility demand.

2.2 Electrification and Sustainable Public Transport

The global transition toward sustainable mobility has accelerated the adoption of electric public transport systems as part of broader decarbonization strategies [10]. Electrified transit systems contribute to greenhouse-gas reduction, energy diversification, improved urban air quality, and lower operational noise levels [22]. Public bus

electrification has become particularly important because buses operate continuously within dense urban corridors where environmental externalities are highly concentrated.

Electric bus systems offer several operational advantages compared with diesel-based fleets, including lower fuel costs, reduced mechanical maintenance requirements, smoother acceleration, and improved passenger comfort [23]. However, the deployment of electric bus systems also introduces new technical and operational challenges associated with charging infrastructure, battery management, power supply reliability, route optimization, and operational scheduling [24].

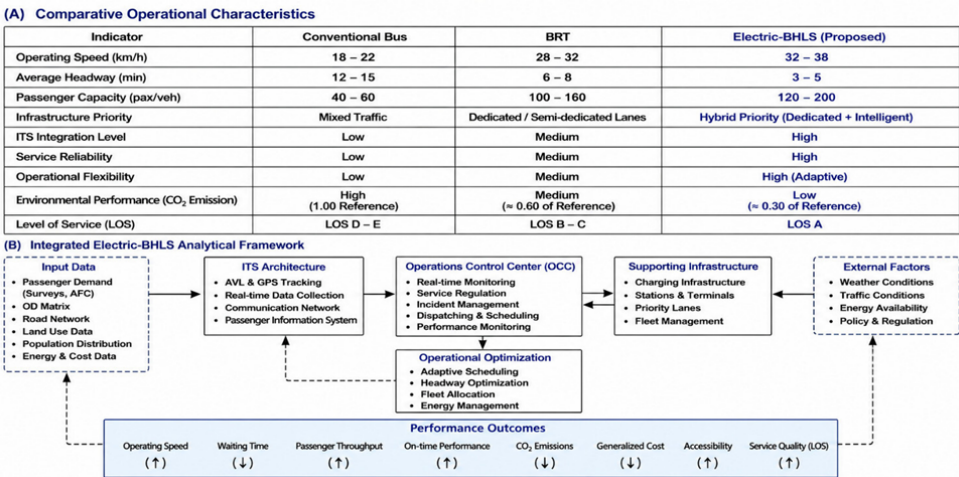


Figure 1. Integrated operational and analytical framework of the proposed next-generation electric bus with high level of service (Electric-BHLS) system

Source: Developed by the authors based on a synthesized literature review and operational characteristics identified in [7, 17, 19, 20], and related studies.

Two principal charging approaches dominate current electric-bus systems: depot charging and opportunity charging. Depot charging relies on overnight charging at centralized facilities, while opportunity charging uses fast pantograph systems located at terminals or intermediate stations to maintain operational continuity throughout the day [25]. Opportunity charging has become increasingly attractive for high-frequency transit corridors because it is projected to reduce battery size requirements and is expected to improve operational flexibility.

Several international case studies demonstrate the growing operational maturity of electric bus systems. Shenzhen developed one of the world’s largest fully electric bus fleets, integrating centralized fleet management, real-time operational monitoring, and large-scale charging infrastructure [26]. Similarly, Electric-BHLS and BRT systems in Amsterdam, Lyon, and Tashkent have demonstrated the feasibility of combining electric propulsion with intelligent operational management and high-capacity service delivery [12, 27].

To strengthen the comparative analytical perspective, selected international electric transit systems were comparatively reviewed using a unified benchmarking framework incorporating operational, environmental, and service-performance indicators relevant to Electric-BHLS deployment. Table 1 presents a comparative benchmarking of selected international electric transit systems.

Nevertheless, many developing countries continue to face substantial barriers to electric transit deployment, including unstable electricity supply, weak institutional coordination, financing limitations, insufficient charging infrastructure, and limited technical expertise [28]. These constraints are particularly significant in Iraq, where transport electrification remains at an early developmental stage.

2.3 Intelligent Transport Systems and Operational Optimization

ITS constitutes a fundamental component of modern high-service transit systems. ITS technologies improve operational reliability, fleet coordination, passenger information management, and traffic-priority control through real-time data integration and digital monitoring systems [29].

Modern BHLS and electric-transit systems increasingly rely on integrated ITS architectures that combine GPS fleet tracking, automated fare collection, adaptive signal priority, CCTV surveillance, passenger-information systems, and centralized operational management [30]. These technologies enable transport operators to optimize service frequency, monitor operational disruptions, improve passenger safety, and reduce variability in travel times.

Operational optimization has similarly become a critical research area in public transport planning. Several studies have developed generalized cost models integrating passenger waiting time, in-vehicle travel time, operator

cost, and environmental externalities into unified optimization frameworks [31]. Such models are particularly relevant for high-frequency transit systems where operational decisions significantly affect both service quality and financial sustainability.

Recent optimization studies have increasingly incorporated environmental objectives alongside operational efficiency. Multi-objective optimization frameworks now commonly integrate emissions reduction, energy consumption, accessibility improvement, and service reliability into public transport evaluation models [32].

Table 1. Comparative benchmarking of selected international electric transit systems

System	Transit Type	Average Operating Speed	Intelligent Transport Systems (ITS) Integration	Charging Strategy	Environmental Performance	Operational Flexibility
Shenzhen	Electric bus network	High	Advanced	Depot Charging	Very High	Moderate
Amsterdam	Next generation electric bus with high level of service (Electric-BHLS)	High	Advanced	Opportunity charging	Very high	High
Lyon	BHLS/Electric transit	Moderate–High	Advanced	Opportunity charging	High	High
Tashkent	Electric-BRT/BHLS	Moderate	Moderate	Mixed charging	Moderate-High	Moderate
Proposed Najaf Corridor	Electric-BHLS	High (projected)	Integrated ITS and Operations Control Center (OCC)	Opportunity charging	High	High

Source: Developed by the authors based on synthesized international transit-system literature and comparative operational analysis from [12, 26, 27], and related studies.

2.4 Regional Accessibility and Urban–Rural Integration

Transport accessibility represents one of the most important determinants of regional development and socio-economic integration [33]. Efficient transport corridors improve labor mobility, facilitate market access, strengthen social connectivity, and support regional economic diversification. In developing regions, transport accessibility is particularly important because infrastructure deficiencies frequently reinforce spatial inequalities between urban and rural communities.

The Najaf–Al-Manathira–Al-Meshkhab corridor represents a mixed urban–rural mobility system characterized by diverse travel purposes, including employment commuting, agricultural logistics, educational access, healthcare movement, and religious tourism. Consequently, corridor performance cannot be evaluated solely through engineering indicators such as speed or capacity, but must also incorporate accessibility, social inclusion, and regional integration dimensions.

GIS-based accessibility analysis has become widely used in transport planning to evaluate service coverage, travel-time accessibility, and connectivity between transport infrastructure and socio-economic activities [34]. Recent studies demonstrate that improved public transport accessibility can significantly enhance employment access, educational opportunities, healthcare connectivity, and regional economic performance [35].

Despite the growing literature on sustainable transport and electric mobility, limited research has integrated Electric-BHLS systems, operational optimization, ITS architecture, and GIS-based regional accessibility analysis within a unified framework, particularly in developing countries and Middle Eastern contexts. This study addresses this gap by developing an integrated Electric-BHLS planning model tailored to the operational, institutional, and infrastructural realities of Iraq.

3 Study Area and Corridor Characteristics

3.1 Regional Context of Najaf Governorate

Najaf Governorate is in central Iraq, approximately 160 km south of Baghdad, and represents one of the country's most important religious, economic, and regional urban centers. The governorate possesses strategic importance due to its role as a global religious destination attracting millions of visitors annually, particularly during major pilgrimage seasons and religious events [36]. In addition to its religious significance, Najaf functions as a regional economic and administrative hub connecting urban, agricultural, and service-oriented activities across central and southern Iraq.

Over the last two decades, Najaf has experienced rapid demographic and spatial expansion driven by population growth, urbanization, economic diversification, and increased tourism activity [37]. This urban growth has generated substantial pressure on existing transport infrastructure, especially along the regional corridors connecting the governorate center with surrounding districts and rural settlements.

The governorate's mobility structure is characterized by highly concentrated travel demand toward the city center, where employment opportunities, educational institutions, healthcare services, administrative functions, and tourism-related activities are located. Consequently, the regional road network experiences significant daily and seasonal traffic fluctuations, particularly during religious events.

3.2 The Najaf–Al-Manathira–Al-Meshkhab Corridor

The Najaf–Al-Manathira–Al-Meshkhab corridor constitutes one of the most important regional transport axes within Najaf Governorate. The corridor extends approximately 27 km from Southern Najaf Terminal through Al-Manathira District toward Al-Meshkhab District, linking dense urban areas with semi-urban and agricultural communities [38].

The corridor performs multiple strategic functions simultaneously:

- Daily passenger mobility is associated with employment, education, and healthcare access.
- Agricultural logistics and freight movement from rural production areas toward urban markets.
- Regional commercial connectivity between the southern districts and Najaf city center.
- Seasonal religious and tourism movements related to pilgrimage activities.

The corridor's mixed urban–rural structure generates highly diversified mobility patterns and fluctuating transport demand throughout the day and across different seasons.

Current transport services along the corridor remain dominated by informal small-capacity minibuses and taxis operating without integrated scheduling systems or centralized operational coordination. The absence of structured mass transit infrastructure has contributed to increasing travel-time variability, declining service reliability, and rising transport costs for users.

The selection of the Najaf–Al-Manathira–Al-Meshkhab corridor as a case-study corridor for Electric-BHLS implementation is supported by several operational and spatial characteristics. The corridor possesses a relatively linear regional structure, concentrated daily mobility demand, and strategically distributed passenger terminals that facilitate corridor-level operational coordination. In addition, the corridor accommodates diverse mobility functions, including urban commuting, agricultural connectivity, educational access, healthcare movement, and seasonal religious mobility. The combination of mixed urban–rural travel demand, existing transport fragmentation, and growing regional accessibility pressures makes the corridor particularly suitable for evaluating the applicability of integrated Electric-BHLS systems under developing-country operational conditions.

3.3 Demographic and Socio-Economic Characteristics

The demographic structure of the corridor reflects substantial population concentration within Najaf city, alongside dispersed rural settlements in Al-Manathira and Al-Meshkhab districts [39]. Population growth rates in the governorate have remained relatively high due to natural increase, internal migration, and urban expansion. The youth population represents a dominant demographic category, generating increasing long-term mobility demand associated with employment, education, and daily commuting activities.

Economically, the corridor combines urban service-sector activities with extensive agricultural production systems. Najaf city functions as the principal commercial and administrative center, while Al-Manathira and Al-Meshkhab represent important agricultural production zones supplying regional food markets [40]. This economic interaction reinforces the importance of reliable and efficient transport connectivity between urban and rural areas.

In addition to daily mobility, religious tourism significantly affects transport demand patterns within the governorate. Seasonal pilgrimage periods generate substantial passenger surges, increasing congestion levels and operational pressure on existing transport systems.

3.4 Existing Passenger Terminal Infrastructure and Operational Conditions

Passenger transport activity along the Najaf–Al-Manathira–Al-Meshkhab corridor is currently organized through three principal passenger terminals located at Southern Najaf, Al-Manathira, and Al-Meshkhab. These terminals constitute the primary operational nodes supporting daily passenger movement, regional connectivity, and local transport distribution across the corridor. Despite their strategic importance, the existing terminals continue to operate under conventional operational structures characterized by limited intelligent management systems, fragmented vehicle dispatching practices, and insufficient passenger-service integration [41].

The Southern Najaf Bus Station represents the largest and most operationally significant terminal along the corridor. The terminal was originally established in 1996 and underwent rehabilitation and reconstruction between 2019 and 2020. The facility occupies approximately 40,000 m² and functions as the principal southern gateway for regional passenger movement within Najaf Governorate. Operationally, the terminal accommodates up to 5,000 vehicles and operates continuously over a 24-hour period through day and night shifts. The terminal infrastructure includes administrative offices, accounting facilities, staff areas, fueling services, vehicle shelters, commercial activities, restaurants, cafés, and maintenance workshops. The station currently serves six internal transport routes and twelve external regional routes connecting Najaf with surrounding districts.

Field observations indicate that the Southern Najaf terminal functions as a major passenger concentration node characterized by high operational activity but limited intelligent operational coordination. Vehicle dispatching remains largely dependent on demand-responsive informal practices rather than fixed scheduling systems, generating substantial fluctuations in waiting times and vehicle occupancy rates during peak periods. In addition, the terminal lacks integrated passenger-information systems, centralized digital fleet management, and multimodal operational coordination [42, 43].

Al-Manathira Bus Station performs an intermediate operational role linking Najaf city with southern agricultural and semi-urban districts. The terminal was established in 2006 on an area of approximately 3,750 m² and occupies a strategically important location connecting multiple regional directions, including Najaf city center, Al-Hira, Al-Meshkhab, and Al-Qadisiyah Governorate. The station primarily serves local and regional passenger movement through minibuses and taxis operating with relatively small passenger capacities.

The operational importance of Al-Manathira terminal derives from its function as a transfer and distribution node within the corridor. However, despite its strategic location, the terminal suffers from limited spatial organization, weak passenger facilities, and insufficient operational control mechanisms. Current operations remain highly dependent on informal vehicle accumulation and demand-based departure patterns, reducing service reliability and increasing operational inefficiency during congestion periods [44].

Al-Meshkhab Bus Station constitutes the southern terminal node of the corridor and supports mobility between agricultural production areas and the regional urban center of Najaf. Established in 2002 on approximately 2,750 m², the terminal is located near the district center and directly connected to the principal regional roadway. The facility includes basic operational infrastructure such as parking areas, passenger waiting shelters, administrative spaces, driver rest facilities, and light maintenance services.

Although the terminal provides essential regional connectivity for rural communities, operational capacity remains relatively limited due to dependence on small-capacity vehicles and the absence of integrated transit-management systems. The terminal currently lacks advanced passenger information systems, digital scheduling mechanisms, and intelligent fleet coordination technologies. Furthermore, the existing infrastructure remains insufficient to accommodate future passenger growth associated with regional urban expansion and increasing mobility demand [45]. Table 2 summarizes the principal operational characteristics, vehicle composition, and passenger capacities of the three major passenger terminals along the corridor.

3.5 Strategic Need for a Next-Generation Electric Bus with High Level of Service System

The transport conditions observed along the corridor indicate the need for a structured high-capacity transit system capable of simultaneously improving operational efficiency, regional accessibility, environmental sustainability, and long-term mobility resilience. Conventional transport solutions based solely on roadway expansion are unlikely to provide sustainable long-term performance due to continued population growth and increasing travel demand [37]. An Electric-BHLS system offers several strategic advantages for the corridor [7, 24, 25]:

- Improved passenger capacity and operational reliability.
- Reduced dependence on informal transport modes.
- Lower environmental emissions and energy consumption.
- Enhanced accessibility between urban and rural communities.
- Integration of ITS-based operational management.
- Greater adaptability compared with heavy BRT systems.

Moreover, the corridor's relatively linear structure, concentrated mobility flows, and strategic regional role make it operationally suitable for phased Electric-BHLS implementation [19, 29]. Figure 2 illustrates the phased

Table 2. Comparative benchmarking of selected international electric transit systems

Terminal	Year Established	Approximate Area (m ²)	Primary Function	Main Vehicle Types	Total Vehicles	Passenger Capacity per Vehicle	Main Operational Characteristics	Key Infrastructure Components	Major Operational Limitations
Southern Najaf Bus Station	1996 (rehabilitated 2019–2020)	40,000	The principal regional gateway connecting Najaf with the southern districts and regional routes	Mini Bus/Taxi	65 Mini Buses/46 Taxis	11/5	24-hour operation; major passenger concentration node; six internal and twelve external routes	Administrative offices, accounting facilities, fueling services, vehicle shelters, workshops, restaurants, cafés, and commercial units	Lack of intelligent operational control, irregular dispatching, absence of digital passenger information systems, congestion during peak periods
AlManathira Bus Station	2006	3,750	Intermediate transfer and distribution node connecting Najaf, AlHira, AlMeshkhab, and AlQadisiyah	Mini Bus/Taxi	55 Mini Buses + 40 Taxis (Najaf route); 53 Mini Buses + 38 Taxis (AlMeshkhab route)	11/5	Local and regional passenger distribution; mixed urban/rural mobility function	Passenger waiting areas, local service facilities, and roadside access connections	Weak spatial organization, limited passenger facilities, informal dispatching practices, and low operational coordination
AlMeshkhab Bus Station	2002	2,750	Southern corridor terminal supporting rural-urban mobility and agricultural connectivity	Mini Bus/Taxi	55 Mini Buses/35 Taxis	11/5	Regional passenger transport toward Al-Manathira and Najaf; agricultural mobility support	Parking yard, administrative offices, driver rest facilities, shaded boarding corridors, light maintenance services	Limited operational capacity, absence of ITS systems, lack of centralized scheduling, inadequate infrastructure for future passenger growth

Source: Developed by the authors based on field surveys, Transportation Department data in Al-Najaf Governorate, and operational observations conducted during 2024.

transition from the existing fragmented transport system toward the proposed integrated Electric-BHLS operational framework, including operational control, fleet modernization, ITS integration, hybrid priority infrastructure, and adaptive service management.

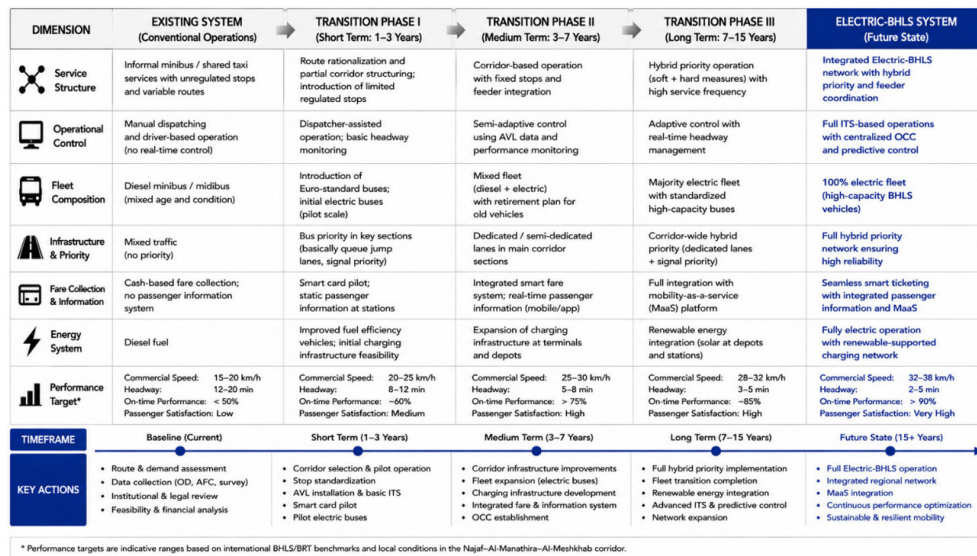


Figure 2. Transition framework toward the proposed next-generation electric bus with high level of service (Electric-BHLS) operational system

4 Proposed Electric Buses with High Level Service System Design and Operational Framework

4.1 Conceptual Framework of the Electric Buses with a High Level Service System

The proposed Electric-BHLS system is designed as an integrated high-capacity regional transit corridor combining electric propulsion, intelligent operational management, selective infrastructure priority, and spatial accessibility optimization [7, 19]. Unlike conventional bus operations, the proposed system emphasizes operational regularity, real-time management, service reliability, environmental sustainability, and multimodal integration within a unified corridor-management structure.

The proposed framework adopts a hybrid operational model that combines dedicated priority segments in high-congestion areas with adaptive mixed-traffic operations in lower-density sections of the corridor [18, 29]. This approach is projected to reduce infrastructure costs while maintaining relatively high operational efficiency and flexibility.

The system integrates five principal operational components:

- Electric articulated bus fleet.
- ITS architecture.
- Opportunity charging infrastructure.
- OCC.
- Integrated station and passenger-information system.

These components operate collectively to achieve improved service regularity, reduced travel time variability, enhanced passenger comfort, and lower environmental impacts.

4.2 Operational Characteristics of the Electric Buses with High Level Service System

One of the principal objectives of the proposed framework is to distinguish the Electric-BHLS system from conventional bus operations and partially segregated BRT systems. While traditional bus systems frequently suffer from irregular scheduling and low operational coordination, the proposed Electric-BHLS model incorporates dynamic operational control supported by ITS technologies and centralized fleet management [30].

The proposed system adopts adaptive service frequency based on demand variation throughout the day. Peak-hour operations are characterized by shorter headways and increased fleet deployment, while off-peak periods apply flexible service intervals to optimize operational efficiency and energy consumption [25]. Table 3 summarizes the proposed operational characteristics of the system.

Table 3. Operational characteristics of the proposed electric bus with high level of service (Electric-BHLS) system

Operational Component	Existing Transport System	Proposed Electric-BHLS System
Vehicle type	Informal minibuses and taxis	Articulated electric buses
Passenger capacity per vehicle	12–18 passengers	120–140 passengers
Operational control	Decentralized and irregular	Centralized Operations Control Center (OCC)-based control
Service frequency	Unscheduled/demand-based	Adaptive scheduled headways
Peak headway	Variable (15–30 min)	5–10 min
Off-peak headway	Variable and inconsistent	10–18 min
Average operating speed	~22 km/h	~35 km/h
Intelligent Transport System (ITS) integration	Limited or absent	Full ITS integration
Global Positioning System (GPS)	No	Yes
Fleet Monitoring	Absent	Real-time digital information
Passenger Information System	Cash-based manual system	Smart ticketing system
Fare collection	No	Selective adaptive priority
Signal priority	Not applicable	Opportunity pantograph charging
Charging infrastructure	Diesel fuel	Electric power
Energy source	High emissions	Reduced CO ₂ emissions
Environmental performance	Limited accessibility	Low-floor accessible boarding
Accessibility standards	Informal roadside stops	Hierarchical smart stations
Station infrastructure	Low and highly variable	High operational reliability
Service reliability	Mixed traffic only	Hybrid priority operation
Traffic integration	Minimal	Real-time operational analytics
Operational data collection		

The operational framework includes:

- Real-time fleet monitoring using GPS systems.
- Adaptive dispatching and headway control.
- Passenger-information systems with real-time updates.
- CCTV-based operational supervision.
- Selective signal-priority mechanisms at critical intersections.
- Smart fare-collection systems.
- Integrated operational performance monitoring [30].

The proposed OCC functions as the central coordination platform for corridor management. The OCC continuously monitors vehicle movement, passenger demand, operational disruptions, charging status, and traffic conditions to maintain service reliability and minimize operational variability.

4.3 Electric Fleet and Charging Infrastructure

The proposed Electric-BHLS corridor utilizes articulated electric buses designed for medium-to-high passenger demand conditions. Articulated buses were selected due to their relatively high carrying capacity, operational flexibility, and suitability for high-frequency transit corridors. The proposed fleet configuration incorporates:

- Low-floor articulated electric buses.
- Average passenger capacity of approximately 120–140 passengers.
- Fast acceleration and regenerative braking systems.
- Integrated smart-monitoring and communication systems.
- Accessibility-compliant boarding infrastructure.

The operational charging strategy adopts opportunity charging through pantograph systems located at major terminals and selected intermediate stations. This approach minimizes operational interruptions while reducing battery-size requirements compared with fully depot-based charging systems [25]. The charging strategy is based on:

- Fast terminal charging during layover periods.
- Supplemental intermediate charging at high-demand stations.
- Overnight depot charging for fleet balancing and maintenance.
- Real-time energy monitoring integrated into the OCC.

The proposed charging infrastructure is designed to accommodate future fleet expansion while maintaining operational continuity under varying demand conditions [19]. Detailed electrical-load simulation and microscale traffic engineering analysis remain beyond the scope of the present corridor-level framework. Nevertheless, the proposed charging and operational structure provides a scalable conceptual and operational basis for future engineering-level implementation studies [7].

4.4 Station Hierarchy and Passenger Facilities

The proposed Electric-BHLS corridor incorporates a hierarchical station structure intended to improve passenger accessibility, operational efficiency, and service integration. Stations are categorized into:

- Primary transfer stations.
- Intermediate operational stations.
- Local accessibility stops.

Primary stations function as multimodal transfer nodes integrating regional bus services, taxi services, feeder routes, and pedestrian access systems [46]. Intermediate stations support corridor operational continuity and passenger distribution, while local stations primarily serve neighborhood accessibility functions.

All stations are designed according to enhanced accessibility and passenger-comfort standards, including:

- Weather-protected waiting areas.
- Real-time passenger-information displays.
- Accessibility ramps and platform-level boarding.
- CCTV surveillance systems.
- Smart-ticketing facilities.
- Solar-assisted lighting systems [34].

The proposed station system aims to improve user experience while simultaneously reducing boarding delays and operational dwell times [30].

The proposed Electric-BHLS corridor adopts a hierarchical station-spacing strategy designed to balance accessibility and operational efficiency under mixed urban–rural corridor conditions. Secondary stops are generally spaced at approximately 600–800 m within dense urban sections to maintain pedestrian accessibility and service coverage, while wider spacing of approximately 1.5–2 km is adopted within semi-urban and rural sections to improve operational speed and reduce unnecessary dwell-time accumulation. Primary transfer stations are positioned at major operational nodes and strategic corridor intersections.

4.5 Intelligent Transport System Integration and Smart Operational Management

ITS integration constitutes a central operational component of the proposed Electric-BHLS framework. The proposed ITS architecture integrates vehicle monitoring, passenger information systems, adaptive signal control, operational optimization, and charging management within a unified digital platform [29].

The ITS system performs several critical operational functions:

- Monitoring vehicle location and speed.
- Managing service regularity and headways.
- Coordinating charging operations.
- Identifying operational disruptions in real time.
- Supporting adaptive traffic-priority mechanisms.

- Collecting operational performance data.
- Improving passenger-information reliability [30].

Adaptive signal-priority systems are proposed at selected intersections with recurrent congestion conditions. These systems allow approaching Electric-BHLS vehicles to receive temporary signal priority, reducing delays and improving corridor travel-time consistency [47].

The ITS framework additionally supports long-term operational planning by generating large-scale mobility datasets that can be used for:

- Demand forecasting.
- Service optimization.
- Maintenance planning.
- Environmental performance evaluation.
- Future network expansion [48].

4.6 Governance and Institutional Framework

The operational success of Electric-BHLS systems depends not only on technical infrastructure but also on effective governance coordination and institutional integration. Fragmented institutional responsibilities frequently represent a major barrier to sustainable transport implementation in developing countries [3, 49].

Accordingly, the study proposes a unified corridor governance authority responsible for:

- Operational supervision.
- Infrastructure coordination.
- Fleet management oversight.
- ITS integration.
- Financial administration.
- Public–private partnership coordination.
- Long-term corridor planning [50].

The proposed governance structure adopts a phased DBOM–PPP framework to improve financial feasibility and reduce long-term operational risks [51]. The governance framework aims to strengthen institutional coordination, improve operational accountability, and support the long-term sustainability of the proposed system.

5 Methodology and Optimization Framework

5.1 Research Design and Analytical Approach

This study adopts an integrated socio-technical and engineering-based research framework to evaluate the feasibility and operational performance of the proposed Electric-BHLS corridor. The methodological structure combines spatial analysis, operational modeling, transport engineering assessment, generalized cost optimization, and sustainability evaluation within a unified analytical framework [19, 24, 31].

The research design is based on the premise that sustainable regional transport systems cannot be evaluated solely through engineering indicators such as speed or capacity, but must simultaneously incorporate operational reliability, environmental performance, accessibility improvement, and institutional feasibility [49, 50]. Consequently, the methodology integrates both quantitative and qualitative analytical dimensions.

Figure 3 presents the integrated methodological, operational, environmental, and governance framework adopted for the proposed Electric-BHLS corridor analysis.

The methodological process consists of five principal analytical stages:

- Spatial and corridor assessment.
- Field-data collection and operational analysis.
- Electric-BHLS system modeling.
- Optimization and scenario evaluation.
- Sustainability and governance assessment.

5.2 Data Collection and Field Surveys

The empirical component of the study is based on field investigations conducted during 2024 along the Najaf–Al-Manathira–Al-Meshkhab corridor. Data collection included:

- Traffic-volume observations.
- GPS-based travel-time measurements.
- Passenger interviews and questionnaires.
- Existing transport-service assessment.
- Corridor infrastructure evaluation.
- Peak and off-peak operational monitoring.

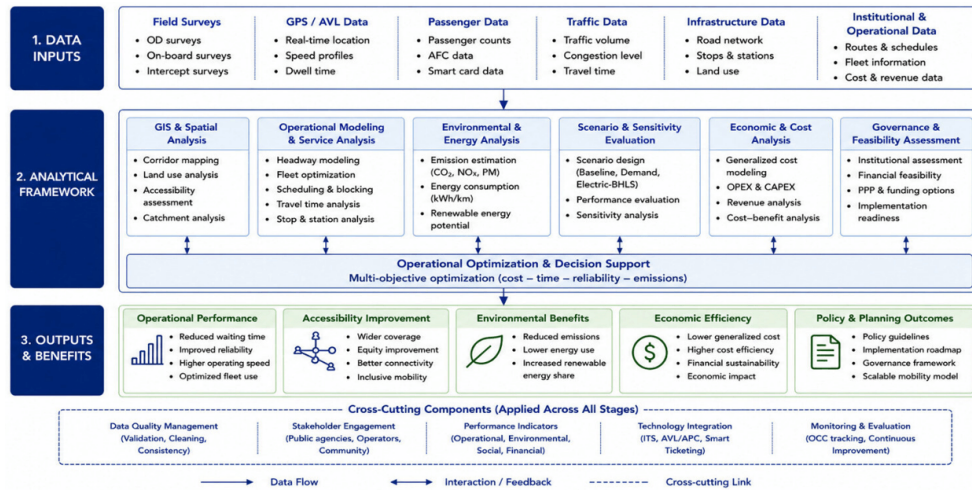


Figure 3. Integrated methodological framework of the proposed electric bus with high level of service (Electric-BHLS) corridor analysis

Field observations were conducted during both weekday and seasonal high-demand periods to capture variations in operational performance and passenger demand [29, 30]. GPS tracking systems were used to measure:

- Average operating speed.
- Intersection delays.
- Corridor travel time.
- Stop dwell time variability.
- Congestion exposure levels [30].

Passenger surveys focused on:

- Waiting time perception.
- Service reliability.
- Accessibility constraints.
- Modal preference.
- Passenger satisfaction.
- Average trip purpose and frequency.

A total of 420 passenger survey responses were collected across the Southern Najaf, Al-Manathira, and Al-Meshkhab terminals during both peak and off-peak operating periods in 2024. The survey sample included daily commuters, students, workers, rural passengers, and inter-district travelers to capture variations in travel behavior and service perception across different user groups [46].

The collected data were subsequently integrated into GIS and operational optimization models for analytical processing [24, 31].

5.3 Geographic Information System-Based Spatial Accessibility Analysis

GIS were used to evaluate spatial accessibility, service coverage, and corridor connectivity. GIS analysis focused on measuring accessibility improvements associated with the proposed Electric-BHLS corridor under different operational scenarios [33, 34].

The spatial analysis incorporated:

- Population-distribution mapping.
- Station catchment-area analysis.
- Service-coverage assessment.
- Accessibility to healthcare and educational facilities.
- Urban–rural connectivity evaluation.
- Corridor interaction analysis [33].

Station catchment areas were evaluated using buffer-distance analysis and travel-time accessibility thresholds. Accessibility indicators were measured before and after implementation of the proposed Electric-BHLS system to estimate potential improvements in regional connectivity [31]. The proposed station hierarchy and spacing strategy were calibrated according to corridor density, pedestrian accessibility thresholds, and operational service-efficiency considerations.

The GIS framework additionally supported the identification of:

- High-demand activity zones.
- Operational bottlenecks.
- Strategic transfer locations.
- Priority charging-station locations.
- Potential multimodal integration points [29].

Future studies may incorporate OD-based accessibility modeling and socio-spatial accessibility differentiation among demographic groups [52].

5.4 Operational and Generalized Cost Modeling

The generalized cost framework integrates passenger costs, operational efficiency, travel time, waiting time, and environmental externalities into a unified analytical structure [31, 53].

$$GC = C_o + \text{VOT} (T_t + T_w) + C_e \quad (1)$$

where, GC = generalized transport cost, C_o = direct operational cost, VOT = value of passenger travel time, T_t = in-vehicle travel time, T_w = passenger waiting time, and C_e = environmental externality cost.

The generalized cost analysis was calibrated using observed corridor travel conditions collected during field surveys in 2024 [30, 46]. Passenger waiting-time values, operational speeds, and dwell-time assumptions were derived from GPS-based measurements and terminal observations conducted during both normal and peak-demand periods. Passenger demand projections were estimated based on existing terminal activity levels, observed vehicle occupancy rates, and projected service-frequency improvements under the proposed Electric-BHLS framework [19, 53].

The operational model evaluates several key performance indicators (KPIs), including:

- Average operating speed.
- Passenger waiting time.
- Corridor travel time.
- Fleet utilization rate.
- Passenger carrying capacity.
- Energy consumption.
- Service reliability.
- CO₂ emissions.

Peak and off-peak operational scenarios were evaluated separately to analyze demand variation and operational flexibility [24, 53].

5.5 Environmental and Energy Assessment

Environmental evaluation focused on estimating the potential reduction in greenhouse-gas emissions and energy consumption associated with replacing conventional diesel-based transport services with Electric-BHLS operations [11, 25].

The environmental analysis incorporated:

- Diesel fuel-consumption estimation.
- Electric energy-consumption modeling.
- CO₂ emission-factor analysis.
- Passenger-capacity efficiency comparison.
- Energy-demand assessment.

The total corridor emissions were estimated using standard emission-factor methodologies based on fleet activity, travel distance, and energy source characteristics [11, 54].

Emission factors for diesel-based operations were derived from standard urban-bus emission references commonly applied in developing-country transport studies, while electricity-consumption assumptions for the proposed electric fleet were estimated using average articulated electric-bus energy-consumption rates reported in recent international electric-transit literature [24, 25]. The environmental assessment focused primarily on comparative corridor-level performance rather than full life-cycle emission accounting.

$$E = EF \times D \times N \quad (2)$$

where, E = total emissions, EF = emission factor, D = operational distance, and N = number of vehicle trips.

Charging-energy requirements and fleet electricity demand were similarly estimated using operational scheduling and vehicle-energy-consumption assumptions.

The operational evaluation of the proposed Electric-BHLS corridor is based on a set of analytical variables, scenario assumptions, and field-calibrated operational parameters derived from surveys, GPS measurements, GIS analysis, and corridor observations conducted during 2024. Table 4 summarizes the principal variables, parameter

assumptions, units, baseline values, and primary data sources used in the operational and sensitivity-analysis framework.

Table 4. Operational variables, modeling parameters, and data sources used in the Electric-BHLS scenario analysis

Variable	Symbol	Unit	Baseline Value	Proposed Electric-BHLS Value	Data Source
Average operating speed	(V_o)	km/h	22	35	Authors' GPS field survey conducted in 2024
Passenger waiting time	(T_w)	min	18	7	Passenger survey & terminal observation
Corridor travel time	(T_t)	min	74	46	GPS tracking analysis
Daily passenger capacity	(P_c)	passengers/day	15,000	36,000	Operational demand estimation
CO ₂ emissions	(E_c)	% reduction	Baseline	-29%	Emission-factor estimation
Urban accessibility	(A_u)	% improvement	Baseline	+24%	GIS accessibility analysis
Rural accessibility	(A_r)	% improvement	Baseline	+38%	GIS accessibility analysis
Fleet size	(F_s)	vehicles	Informal fleet	20–30 buses	Scenario assumptions
Charging duration	(C_d)	min	N/A	15–25	Operational scenario modeling
Headway	(H)	min	15–30	5–10	Operational scheduling model

Note: Electric-BHLS = electric bus with high level of service; GPS = Global Positioning System; GIS = Geographic Information System. The presented variables and parameter values represent scenario-based operational assumptions and projected analytical estimates derived from field surveys, GPS measurements, passenger interviews, GIS accessibility analysis, and corridor operational observations conducted during 2024.

5.6 Scenario Analysis and Sensitivity Evaluation

To improve analytical reliability and evaluate the operational robustness of the proposed Electric-BHLS corridor, the study examines multiple operational scenarios based on variations in passenger demand, fleet size, service frequency, charging duration, electricity cost, and peak-period congestion conditions. Sensitivity analysis was conducted to assess how changes in these operational parameters may influence key performance indicators, including waiting time, operational reliability, generalized transport cost, and service efficiency.

The scenario-based framework includes:

- Existing fragmented transport conditions (baseline scenario);
- Moderate-demand Electric-BHLS operational conditions;
- High-demand operational scenarios;
- Expanded future operational conditions associated with long-term corridor growth.

The selected scenarios were designed to represent realistic variations in corridor demand, congestion intensity, and operational conditions expected under regional mobility growth.

The comparative scenario analysis, therefore, aims to support long-term operational planning under uncertainty conditions and improve the analytical robustness of the proposed framework [53].

The operational parameters summarized in Table 5 provide the analytical basis for evaluating the projected performance of the proposed Electric-BHLS corridor under varying operational conditions. The comparative scenario framework was used to estimate changes in waiting time, operational reliability, generalized transport cost, environmental performance, and regional accessibility across different demand and congestion conditions. The outputs generated from these scenarios were subsequently integrated into the operational, environmental, and accessibility assessments presented in Section 6.

The comparative scenario analysis indicates that operational performance is strongly influenced by fleet size, charging duration, passenger-demand variability, and congestion conditions. Increasing fleet availability improves service regularity and reduces passenger waiting time under high-demand conditions, while extended charging

duration may reduce operational efficiency during peak periods if reserve fleet capacity remains limited. The analysis further suggests that hybrid priority measures and ITS-supported operational coordination remain essential for maintaining schedule reliability and corridor performance stability under congestion-sensitive conditions. Scenario B demonstrates the highest projected operational efficiency due to increased fleet capacity, coordinated dispatching strategies, and improved service frequency.

Table 5. Comparative operational scenarios used in the electric bus with high level of service (Electric-BHLS) sensitivity analysis

Scenario	Passenger Demand	Fleet Size	Charging Duration	Estimated Waiting Time	Operating Cost Efficiency	Operational Reliability
Existing conditions	Current fragmented demand	Informal vehicles	N/A	High	Low	Low
Scenario A	Moderate demand growth	20 electric buses	15 min	Moderate	Improved	Moderate
Scenario B	High passenger demand	30 electric buses	20 min	Low	Optimized	High
Scenario C	Congested operational conditions	30 electric buses	25 min	Moderate	Reduced	Moderate

Nevertheless, the present analytical framework does not fully incorporate dynamic passenger behavioral responses, charging-conflict simulation, or elasticity-based demand forecasting due to data limitations and the pre-operational nature of the proposed system. Future studies may therefore integrate advanced dynamic simulation models, behavioral-demand analysis, and real-time operational variability assessment to improve predictive accuracy and operational realism [48].

5.7 Governance and Feasibility Assessment

In addition to technical analysis, the methodology incorporates governance and implementation-feasibility assessment. This component evaluates:

- Institutional coordination requirements.
- Operational management capacity.
- Financial sustainability.
- Public–private partnership feasibility.
- Infrastructure implementation constraints.
- Electricity-supply reliability.
- Long-term maintenance requirements.

The governance assessment is particularly important within the Iraqi context, where fragmented institutional responsibilities and infrastructure limitations significantly influence transport-system implementation capacity [3, 49, 50].

The integrated methodological framework, therefore, combines engineering analysis, operational optimization, environmental assessment, spatial accessibility evaluation, and institutional analysis within a unified Electric-BHLS planning structure suitable for developing-country regional corridors.

6 Results and Discussion

The findings presented in this section represent simulation-based and scenario-based estimates derived from comparative operational modeling, GIS accessibility analysis, and projected Electric-BHLS operational assumptions. Since the proposed corridor system has not yet been implemented in practice, the reported results should not be interpreted as validated operational observations but rather as analytical estimates intended to evaluate the potential performance and regional mobility implications of the proposed framework. The presented results, therefore, represent projected performance conditions derived from comparative operational analysis, engineering assumptions, GIS-based accessibility assessment, and structured scenario evaluation. This section presents the operational, spatial, environmental, and governance outcomes of the proposed Electric-BHLS framework. The analysis integrates operational simulations, GIS-based accessibility assessment, generalized cost evaluation, and sustainability indicators to examine the corridor-level impacts of the proposed system under different operating conditions.

6.1 Operational Performance of the Proposed Electric Bus with High Level of Service System

The analytical results indicate that the proposed Electric-BHLS system is expected to substantially improve operational performance compared with the existing corridor transport conditions. Current transport operations along the Najaf–Al-Manathira–Al-Meshkhab corridor are characterized by irregular service frequency, mixed-traffic exposure, high waiting times, and low operating speeds. The implementation of the proposed Electric-BHLS framework is expected to significantly improve service regularity, operational coordination, and passenger mobility efficiency [19, 29].

The reported operational improvements represent scenario-based simulation outcomes calibrated using observed corridor operating conditions, field measurements, and comparative Electric-BHLS operational assumptions derived from previous international high-service transit studies [24, 53]. The results, therefore, represent analytically estimated performance improvements under structured operational management conditions rather than direct post-implementation measurements.

The operational simulations demonstrate that average operating speed increases from approximately 22 km/h under existing conditions to approximately 35 km/h after implementation of the proposed system.

This improvement is primarily associated with:

- Adaptive operational scheduling.
- Selective priority-lane implementation.
- ITS-supported signal management.
- Reduced dwell-time variability.
- Centralized operational control [19, 30].

Passenger waiting time similarly declines substantially due to adaptive headway management and improved dispatching coordination [48]. The analysis indicates that the average passenger waiting time decreases by approximately 61% during normal operating conditions.

Table 6 summarizes the principal operational improvements associated with the proposed system.

The operational results further indicate improved fleet utilization efficiency and greater service reliability during both peak and off-peak periods. Unlike the existing fragmented transport system, the proposed Electric-BHLS framework maintains more stable operational headways and reduced travel-time variability under fluctuating demand conditions [31, 53].

Figure 4 presents the comparative operational-performance indicators before and after implementation of the Electric-BHLS system.

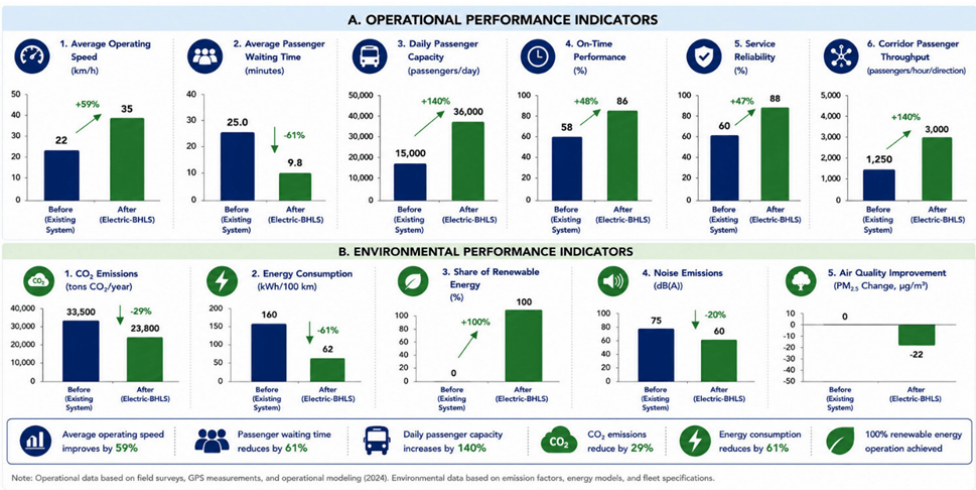


Figure 4. Comparative operational performance indicators

Beyond operational efficiency, the proposed system also generates substantial improvements in regional accessibility and spatial connectivity.

6.2 Accessibility and Spatial Connectivity Outcomes

GIS-based accessibility scenario analysis suggests that the proposed Electric-BHLS corridor significantly is expected to improve regional connectivity and service accessibility across both urban and rural areas. The implementation of structured station hierarchy and improved operational frequency expands effective service coverage and is projected to reduce travel barriers for surrounding communities [33, 52].

The GIS-based accessibility analysis indicates several significant spatial improvements, including:

Table 6. Comparative operational performance results

Performance Indicator	Existing Transport System	Proposed Electric-BHLS System	Improvement
Average operating speed	22 km/h	35 km/h	+59%
Average passenger waiting time	18 min	7 min	-61%
Corridor travel time	74 min	46 min	-38%
Daily passenger capacity	~15,000 passengers/day	~36,000 passengers/day	+140%
Service reliability	Low	High	Significant improvement
Fleet utilization efficiency	Moderate	High	Improved operational efficiency
CO ₂ emissions	High	Reduced	-29%
Fuel/energy consumption efficiency	Low	High	Improved efficiency
Accessibility coverage (urban areas)	Baseline	Expanded coverage	+24%
Accessibility coverage (rural areas)	Limited	Improved regional access	+38%
Passenger information availability	Minimal	Real-time digital system	Major improvement
Operational monitoring	Fragmented	OCC-based centralized monitoring	Full integration
Traffic delay exposure	High	Moderate	Reduced congestion exposure
Fare collection efficiency	Manual cash system	Smart ticketing	Improved boarding efficiency
Environmental sustainability	Weak	High	Significant improvement
Multimodal connectivity	Limited	Integrated transfer system	Improved integration
Peak-hour service stability	Irregular	Stable adaptive headways	Major improvement
Passenger comfort level	Low	Enhanced	Significant improvement

Note: Electric-BHLS = electric bus with high level of service; OCC = Operations Control Center.

- Approximately 24% improvement in urban accessibility indicators.
- Approximately 38% improvement in rural accessibility levels.
- Improved access to educational and healthcare facilities.
- Enhanced connectivity between agricultural districts and urban service centers.

The most substantial accessibility improvements are observed in peripheral and semi-rural zones previously characterized by weak transport-service reliability and limited mobility opportunities [52]. Figure 5 illustrates the spatial accessibility changes associated with the proposed corridor implementation.

The GIS results additionally demonstrate that strategic station placement is expected to significantly improve multimodal connectivity and corridor interaction efficiency [46]. The proposed station hierarchy is projected to reduce excessive concentration of passenger movement within central Najaf while strengthening regional transport integration along the corridor [33].

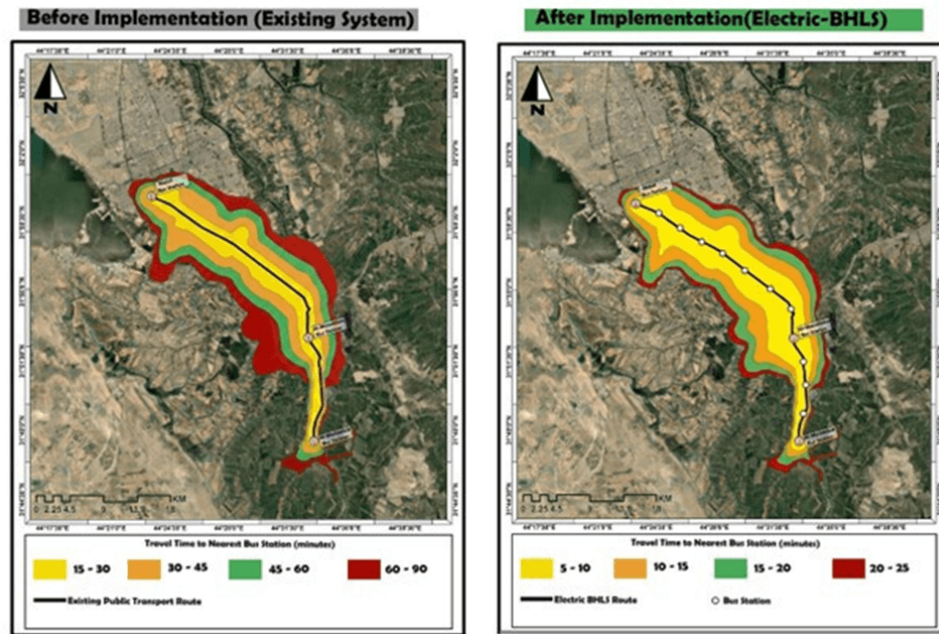


Figure 5. Spatial accessibility improvements after electric bus with high level of service (Electric-BHLS) implementation

6.3 Environmental and Energy Performance

The environmental analysis indicates that replacing conventional diesel-based transport operations with Electric-BHLS services is projected to substantially reduce corridor emissions and energy consumption intensity [11].

The estimated results demonstrate:

- Approximately 29% reduction in CO₂ emissions.
- Lower fuel-consumption dependency.
- Reduced local air-pollution exposure.
- Improved energy efficiency per passenger-kilometer.

The environmental benefits primarily result from:

- Fleet electrification.
- Higher passenger-carrying efficiency.
- Reduced congestion exposure.
- Improved operational regularity.
- Lower idle-time duration [53].

The analysis further indicates that opportunity charging is expected to improve operational continuity while reducing the need for oversized battery systems. However, long-term environmental performance remains partially dependent on electricity-generation sources and grid reliability within Iraq.

6.4 Generalized Cost and Economic Efficiency

The generalized cost analysis scenario analysis suggests significant economic efficiency improvements associated with the proposed Electric-BHLS system. The integrated optimization model indicates approximately 27% reduction in generalized transport cost relative to existing transport conditions. The reduction in generalized cost is primarily associated with:

- Reduced passenger waiting time.
- Faster corridor travel speeds.
- Lower operational variability.
- Improved fleet efficiency.
- Reduced environmental externalities [31, 53].

Figure 6 illustrates the distribution of generalized transport-cost components under existing and proposed operational conditions.

From an infrastructure perspective, the proposed Electric-BHLS framework remains more financially feasible than full-scale BRT or rail-based systems because it combines selective infrastructure priority with adaptive mixed-traffic operation [19, 20].

6.5 Operational Reliability and Intelligent Transport System Contribution

One of the most significant findings of the study is the critical role of ITS integration in improving operational stability and service reliability [29]. The proposed ITS framework is expected to substantially improve:

- Headway consistency.
- Dispatching coordination.
- Passenger-information reliability.
- Charging synchronization.
- Traffic-priority management.

Adaptive operational control is projected to reduce the operational instability commonly associated with conventional bus systems operating under mixed-traffic conditions [48]. The OCC is additionally expected to improve corridor resilience by enabling rapid operational adjustment during congestion events, peak demand periods, or operational disruptions.

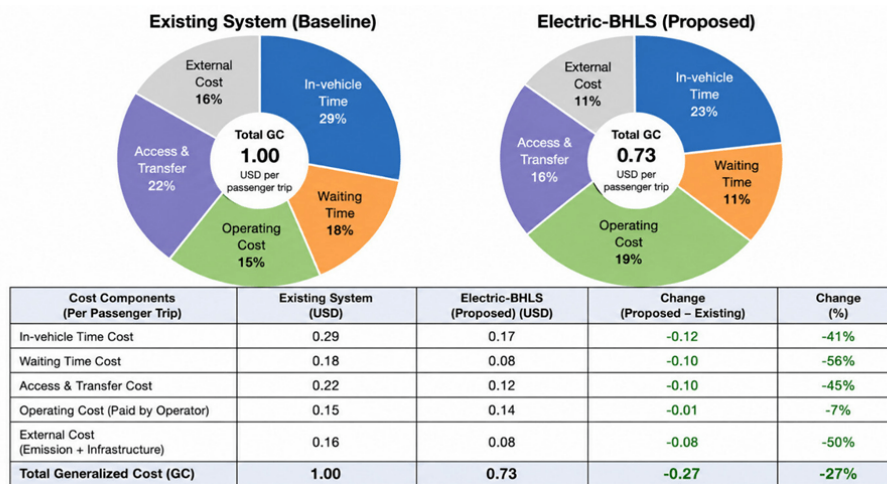


Figure 6. Generalized cost structure before and after electric bus with high level of service (Electric-BHLS) implementation

The findings suggest that ITS integration constitutes one of the principal distinguishing characteristics separating Electric-BHLS systems from conventional bus operations [19].

6.6 Governance and Implementation Challenges

Despite the positive operational and environmental results, the study identifies several implementation challenges that may affect long-term feasibility under the Iraqi context [3, 49].

The principal implementation constraints include:

- Fragmented institutional responsibilities.
- Electricity-supply instability.
- Financing limitations.
- Weak technical maintenance capacity.
- Limited ITS operational experience.
- Traffic-management coordination difficulties.

The governance analysis indicates that successful implementation requires strong institutional coordination supported by a unified operational authority and a phased implementation strategy. The PPP mechanisms may improve financial feasibility; however, long-term operational sustainability remains dependent on stable governance structures and continuous technical capacity development [51].

The results additionally indicate that phased implementation provides a more realistic strategy than full immediate deployment, particularly under infrastructure and financing constraints.

6.7 Comparative Discussion and Research Implications

Compared with previous Electric-BHLS and BRT studies, the proposed framework integrates operational optimization, ITS-supported management, GIS-based accessibility analysis, governance evaluation, environmental assessment, and regional development considerations within a unified corridor-scale planning model [7, 24].

The study therefore contributes to the growing literature on sustainable regional transit systems by proposing a transferable Electric-BHLS framework suitable for medium-sized and developing-country corridors characterized by mixed urban–rural mobility structures [19].

The findings further suggest that Electric-BHLS systems can provide an effective intermediate solution between low-capacity informal transport systems and highly capital-intensive rail or fully segregated BRT infrastructure. This flexibility may be particularly valuable for rapidly urbanizing regions facing financial and institutional constraints [20].

Overall, the results demonstrate that the proposed Electric-BHLS framework provides substantial operational, environmental, and accessibility advantages compared with the existing fragmented transport conditions along the Najaf–Al-Manathira–Al-Meshkhab corridor. The integration of ITS-supported operational management, adaptive service scheduling, electric propulsion, and structured station hierarchy significantly is expected to improve corridor efficiency, regional connectivity, and service reliability. In addition, the findings indicate that the proposed framework offers a financially and operationally adaptable solution suitable for developing-country regional corridors characterized by infrastructure limitations and institutional fragmentation. These outcomes highlight the broader potential of Electric-BHLS systems to support sustainable regional mobility transition and long-term transport modernization within emerging urban and regional contexts [19, 24, 29].

6.8 Quantitative Scenario and Sensitivity Analysis

To evaluate the operational robustness of the proposed Electric-BHLS corridor under varying conditions, a comparative scenario and sensitivity analysis were conducted. The analysis examines the influence of key operational variables—including passenger demand, fleet size, charging duration, congestion intensity, and electricity cost—on system performance indicators such as waiting time, operational efficiency, generalized transport cost, and environmental performance. Since the corridor has not yet been implemented, the following results represent scenario-based operational estimates derived from comparative modeling assumptions.

The comparative scenario and sensitivity-analysis results are summarized in Table 7, which presents the projected operational performance of the proposed Electric-BHLS system under different demand, fleet, congestion, and electricity-cost conditions.

Table 7. Comparative operational scenarios and sensitivity analysis outcomes for the proposed electric bus with high level of service (Electric-BHLS) system

Scenario	Passenger Demand	Fleet Size	Average Waiting Time (min)	Operating Speed (km/h)	Estimated CO ₂ Reduction	Accessibility Impact
Baseline existing system	Existing demand	14 buses	18	22	Baseline	Existing coverage
Base Electric-BHLS scenario	100%	20 buses	7	35	29% reduction	+24% urban/+38% rural
High-demand scenario	130%	20 buses	11	30	24% reduction	Moderate congestion impact
Expanded fleet scenario	130%	28 buses	6	34	27% reduction	Improved regional coverage
High-congestion scenario	100%	20 buses	9	27	22% reduction	Reduced operational efficiency
Increased electricity cost scenario	100%	20 buses	7	35	29% reduction	Slight operational-cost increase

7 Conclusion and Policy Recommendations

This study presents a conceptual and scenario-based integrated planning and operational framework for an Electric-BHLS corridor along the Najaf–Al-Manathira–Al-Meshkhab regional axis in Iraq. The proposed framework integrates electric mobility, ITS, generalized cost optimization, GIS-based accessibility analysis, and governance assessment within a unified regional transport model suitable for infrastructure-constrained regional corridors.

Since the proposed Electric-BHLS system has not yet been implemented, the reported operational, environmental, and accessibility outcomes represent analytical and scenario-based projections derived from comparative modeling assumptions rather than validated post-implementation performance observations.

Simulation-based analysis suggests that the proposed Electric-BHLS system could substantially improve operational performance, regional accessibility, and environmental sustainability compared with the existing fragmented transport conditions currently dominating the corridor. The projected operational improvements are associated with adaptive operational scheduling, centralized operational control, selective priority infrastructure, and ITS-supported management, which collectively improve service reliability, travel efficiency, and corridor operational stability. The analysis additionally indicates projected reductions in passenger waiting time and travel-time variability, together with improvements in average operating speed and passenger-carrying capacity.

The GIS-based accessibility assessment further indicates that the proposed framework may strengthen regional connectivity between urban, semi-urban, and rural areas while improving access to employment centers, healthcare services, educational facilities, and commercial activities. The structured station hierarchy and optimized corridor operations are also expected to reduce mobility inequalities affecting peripheral and underserved regional communities.

From an environmental perspective, the proposed Electric-BHLS framework supports sustainable mobility objectives through projected reductions in greenhouse gas emissions, lower fossil-fuel dependency, and improved passenger-capacity efficiency. The integration of electric articulated buses and opportunity-charging systems additionally supports long-term operational sustainability relative to conventional diesel-based transport operations. The analysis further suggests that Electric-BHLS systems may provide a financially and operationally adaptable intermediate solution between low-capacity informal transport systems and highly capital-intensive rail or fully segregated BRT infrastructure. This flexibility is particularly important for emerging regional corridors characterized by infrastructure limitations, institutional fragmentation, and constrained public-investment capacity.

Despite these projected benefits, several implementation challenges remain significant within the Iraqi context. These include electricity-supply instability, fragmented institutional responsibilities, financing limitations, limited ITS operational experience, and long-term maintenance requirements associated with electric transit systems. Accordingly, successful implementation will likely depend on phased deployment strategies, strong institutional coordination, and continuous technical-capacity development.

Based on the findings, the study proposes several policy recommendations:

- Establishment of a unified regional transport authority responsible for corridor management and operational integration;
- Adoption of phased Electric-BHLS implementation strategies prioritizing high-demand corridor segments;
- Expansion of ITS infrastructure and smart operational-management systems;
- Development of dedicated financing mechanisms and PPP frameworks;
- Integration of sustainable transport planning into regional development strategies;
- Strengthening technical training and long-term operational-maintenance capacity for electric transit systems.

The study contributes to the growing literature on sustainable regional mobility by proposing an integrated Electric-BHLS framework suitable for medium-sized and rapidly urbanizing regional corridors in developing countries. Unlike many previous studies focused primarily on vehicle technology or conventional BRT systems, the present research combines operational optimization, accessibility evaluation, environmental assessment, and governance analysis within a unified socio-technical planning framework.

Several limitations should nevertheless be acknowledged. Passenger-demand projections, operational optimization parameters, and environmental calculations were primarily based on pre-operational modeling assumptions and corridor-level estimations rather than full real-world deployment data. Similarly, electricity-generation variability, long-term behavioral modal shifts, and dynamic charging-conflict conditions were evaluated conceptually rather than through real-time operational simulation.

Future research may therefore focus on dynamic operational simulation, real-time charging optimization, passenger behavioral-response modeling, and corridor-scale multimodal integration analysis following future implementation stages of the proposed Electric-BHLS system.

Author Contributions

Conceptualization, S.I.K.A. and S.K.K.; methodology, S.I.K.A.; formal analysis, S.I.K.A.; investigation, S.I.K.A. and S.K.K.; resources, S.K.K.; data curation, S.I.K.A.; visualization, S.I.K.A.; writing—original draft preparation, S.I.K.A.; writing—review and editing, S.I.K.A. and S.K.K.; supervision, S.K.K.; project administration, S.I.K.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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